Exercises PVL LTL

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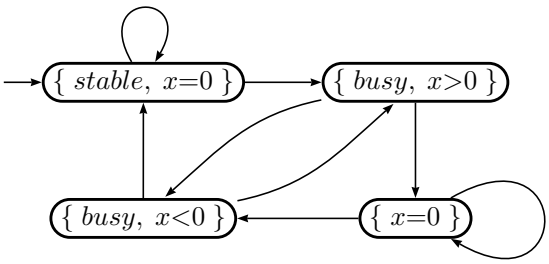
1. Below you see a Kripke structure; let's call it M. Give its explicit definition in terms of a tuple etc (see the formal definition in the slides).



   (a) Why don't we have final states there?

   (b) How is the notion of 'execution' defined for a Kripke structure? And what is an 'abstract execution'?

   (c) Give an execution of M that satisfies the property $\mathbf{X}\ (busy\ \mathbf{U}\ (x{=}0))$. Does M satisfies the property?

   (d) So, given a property Kripke structure M, an (abstract) execution Π, and a property ϕ, and an natural number i, what is the difference between:

   - $M \models \phi$
   - $\Pi \models \phi$
   - $\Pi, i \models \phi$

2. Express the following requirements in LTL. Make the necessary assumptions if you have to; but be reasonable.

   (a) P and Q cannot not use a resource r simultaneously.
   Answer:

   $$\Box\neg(use(P,r) \wedge use(Q,r))$$

   where $use(P,r)$ is a predicate which is true while and as long as P is using r. Importantly note that it does not represent a program call.

   (b) Whenever P requests access to r, eventually it will get the access.
   Answer:

   $$\Box(req(P,r) \rightarrow \Diamond use(P,r))$$

   where $req(P,r)$ is a predicate which is true while and as long as P is requesting for r.

(c) Whenever P requests access to r, eventually it will get the access; but only if P persists on maintaining the request.

Answer:

$$\Box(req(P,r) \ \rightarrow \ (req(P,r) \ \ \mathbf{U} \ \ \neg req(P,r) \vee use(P,r)))$$

(d) P cannot access r without first requesting it; and it cannot do so (make a request) without first releasing r (if it was busy using r).

Answer:

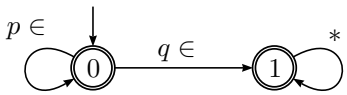
$$\Box((\neg req(P,r) \wedge \neg use(P,r)) \ \mathbf{W} \ (req(P,r) \wedge \neg use(P,r)))$$

$$\Box((use(P,r) \wedge \neg req(P,r)) \ \mathbf{W} \ (\neg use(P,r) \wedge \neg req(P,r)))$$

3. Construct Buchi automata representing the following LTL formulas:

(a) $p \ \mathbf{W} \ q$, where p, q are atomic propositions.

Answer:



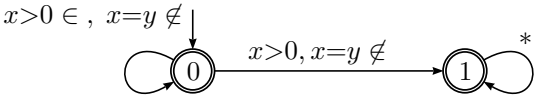
Where the above is a standard Buchi with both states accepting.

(b) $\neg(x{>}0 \ \mathbf{U} \ x{=}y)$

Answer: Note that $\neg(p \ \mathbf{U} \ q) \ = \ (p \wedge \neg q) \ \mathbf{W} \ (\neg p \wedge \neg q)$. So the above property is equivalent to:

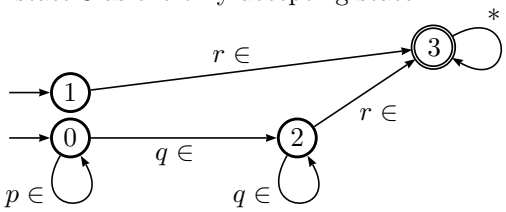
$$(x{>}0 \wedge \neg x{=}y) \ \mathbf{W} \ (\neg x{>}0 \wedge \neg x{=}y)$$

This results in the following standard Buchi automaton. Both states are accepting.



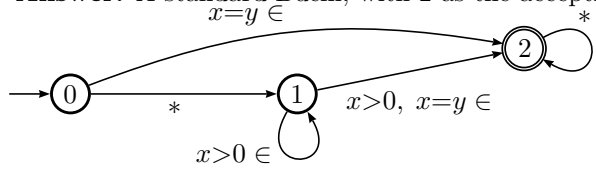
(c) $p \ \mathbf{U} \ (q \ \mathbf{U} \ r)$, where p, q are atomic propositions.

Answer: The following standard Buchi with $\{0, 1\}$ as initial states, and state 3 as the only accepting state.



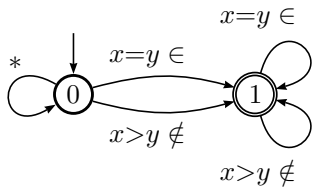
(d) $(\mathbf{X} \ x{>}0) \ \mathbf{U} \ x{=}y$

Answer: A standard Buchi, with 2 as the accepting state.



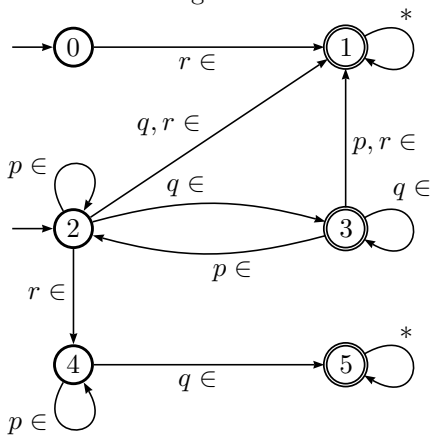
(e) $\Diamond\Box(x{>}0 \rightarrow x{=}y)$

Answer: Notice first that $x{>}0 \rightarrow x{=}y$ can also be written as $\neg(x{>}0) \vee x{=}y$. Below is a standard Buchi with 1 as the accepting state.



(f) $(p \mathbf{U} q) \mathbf{W} r$

Answer: Using this standard Buchi, with $\{0, 2\}$ as the initial states:



With $F = \{1, 3, 5\}$ as the accepting states. Accepting via 1 describes executions whose prefix repeatedly satisfy $p \mathbf{U} q$, zero or more times, and ends up in q (in the case of at least one time$p \mathbf{U} q$); and then it is followed up with r.

Accepting via 3 describes the scenario of executions that remain in $p \mathbf{U} q$ forever, without ever to go over to r.

Finally, accepting via 5 describes executions whose prefix repeatedly satisfy $p \mathbf{U} q$, zero or more times, and then they go over to r; thus still owing one future q, but this future q is met (after r).